UID EX 6

NAME: M AKASH **Roll No:240701025**

RAD (Rapid Application Development) is a software development methodology that emphasizes **quick development and iteration** of prototypes over strict planning and long development cycles. It's ideal for projects where requirements evolve through user feedback and where fast delivery is important.

Key Characteristics of RAD:

- Focus on **prototypes** instead of extensive documentation
- **User involvement** throughout the development process
- **Fast development cycles** with frequent iterations
- **Reusable components** to reduce effort
- Best suited for **UI/UX-intensive projects** like web and mobile apps

<u>Phase 1</u>: Requirements Planning

This is where the **project goals, system requirements, and user needs** are identified. Stakeholders, users, and designers collaborate to decide what needs to be built.

1) Identify Key Features

Navigation:

- Home
- Product Details
- Product Categories
- Cart
- Product Listings
- Order Confirmation

User Actions: Browsing, Searching, Adding to Cart, Checkout, Tracking Orders

2) Create a Requirements Document:

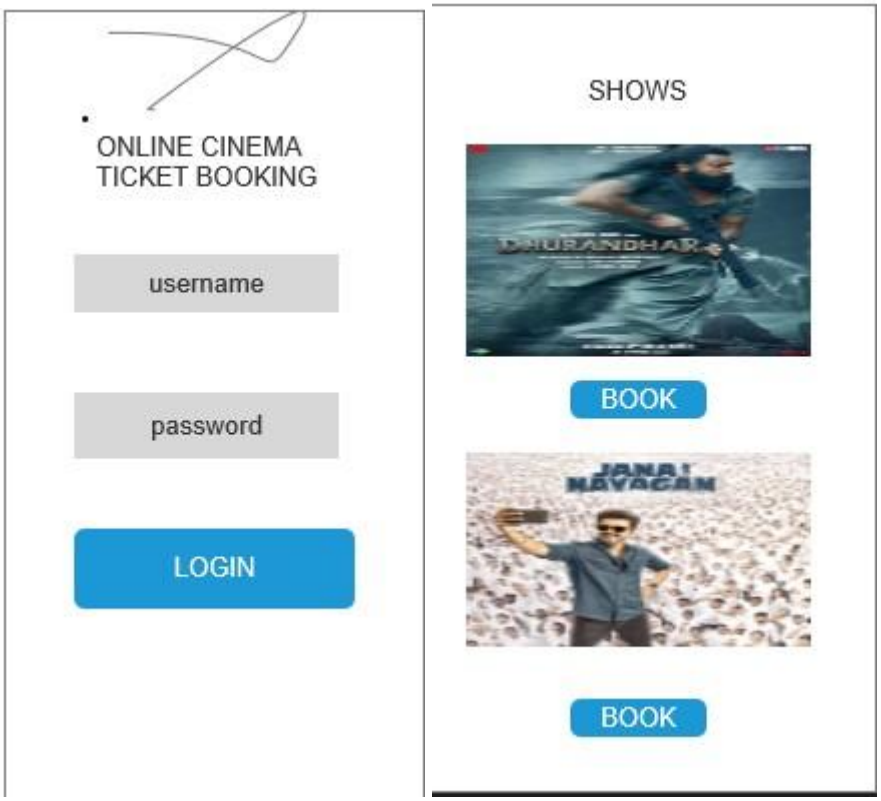
1. View list of product categories
2. View products under each category
3. View detailed information about a product
4. View and edit cart
5. Checkout and enter user/shipping/payment details
6. View order confirmation message

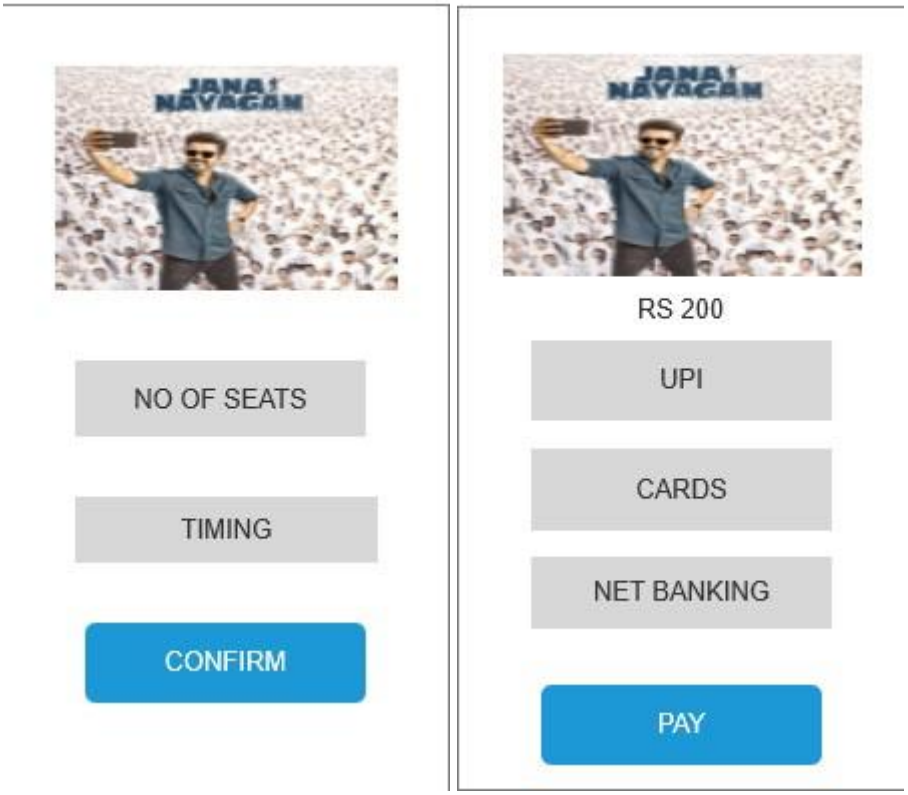
<u>Phase 2</u>: User Design

Designers and users collaborate to **create wireframes and prototypes**. The idea is to show how the application will look and behave before any actual coding starts.

1) **Tools Used: Axure**: Used to design the wireframes and UI prototype.
2) **Wireframes Designed:**
   - **Home Page:** Highlights featured products and links to categories. • **Product Categories:** Lists all product categories to explore.
   - **Product Listings:** Shows products under a selected category.
   - **Product Details:** Displays product info with "Add to Cart" option. •
   Cart: Shows added items with price and quantity details. •
   Checkout: Collects shipping and payment information. •
   Order Confirmation: Confirms successful order placement.

Designed **interactions and navigation** using Axure's Prototype mode Reused components like headers/footers using **Components** Annotated elements for clarity (e.g., what a button does, purpose of sections)





<u>Phase 3</u>: Construction

This is where the **actual development happens**, but in RAD, it is tightly coupled with ongoing testing and feedback. You refine the prototype based on user responses.

1) Interactive Prototypes:

- Created using Axure's **prototype linking** features.
- Functional paths designed:
    - From **Home → Product Categories → Product Listings → Product Details → Add to Cart → Checkout → Order Confirmation**

2) Dynamic Elements:

- Simulated transitions and modals for adding to cart and checking out. • Screens simulate real-time behavior such as navigating between product listings and cart.

3) Testing and Iteration:

- Prototype previewed with **"Present"** mode in Figma.
- Collected feedback from stakeholders (assumed for documentation). • Iterated based on visual hierarchy, accessibility, and flow suggestions.

<u>Phase 4</u>: Cutover

In this final phase, the system goes live. This includes **finalizing the design**, **training users**, and **preparing documentation** for developers.

1) Finalization and Export:

- Final design prototype shared through Figma link.
- Ready for hand-off to development team.
- Could be exported to **HTML/CSS** using plugins like Figma-to-HTML or further tools if needed.

2) User Training & Support:

- Walkthrough of the prototype for stakeholders or development team. • Notes/documentation provided for developers to understand functionality and user flow.
- Easy access to Figma comments for further collaboration and questions.